In case of hetero nuclear diatomic molecules, the covalent radius of individual atom can also be calculated using the internuclear distance ($d_{A-B}$) between two different atoms A and B. The simplest method proposed by Schomaker and Stevenson is as follows.

$$d_{A-B} = r_A + r_B - 0.09\ (\chi_A - \chi_B)$$

where $\chi_A$ and $\chi_B$ are the electronegativities of A and B respectively in Pauling units. Here $\chi_A > \chi_B$ and radius is in Å.

Let us calculate the covalent radius of hydrogen using the experimental $d_{H-Cl}$ value is 1.28 Å and the covalent radius of chlorine is 0.99 Å. In pauling scale the electronegativity of chlorine and hydrogen are 3 and 2.1 respectively.

$$d_{H-Cl} = r_H + r_{Cl} - 0.09\ (\chi_{Cl} - \chi_H)$$

$$1.28 = r_H + 0.99 - 0.09\ (3 - 2.1)$$

$$1.28 = r_H + 0.99 - 0.09\ (0.9)$$

$$1.28 = r_H + 0.99 - 0.081$$

$$1.28 = r_H + 0.909$$

$$\therefore r_H = 1.28 - 0.909 = 0.371\ Å$$

## Metallic radius

It is defined as one-half of the distance between two adjacent metal atoms in the closely packed metallic crystal lattice.

For example, the distance between the adjacent copper atoms in solid copper is 2.56 Å and therefore the metallic radius of copper is

$$\frac{2.56}{2} = 1.28\ Å$$

The metallic radius can be calculated using the unit cell length of the metallic crystal. You will study the detailed calculation procedure in XII standard solid state unit.

## Periodic Trends in Atomic Radius

### Variation in Periods

Atomic radius tends to decrease in a period. As we move from left to right along a period, the valence electrons are added to the same shell. The simultaneous addition of protons to the nucleus, increases the nuclear charge, as well as the electrostatic attractive force between the valence electrons and the nucleus. Therefore atomic radius decreases along a period.

### Effective nuclear charge

In addition to the electrostatic forces of attraction between the nucleus and the electrons, there exists repulsive forces among the electrons. The repulsive force between the inner shell electrons and the valence electrons leads to a decrease in the electrostatic attractive forces acting on the valence electrons by the nucleus. Thus, the inner shell electrons act as a shield between the nucleus and the valence electrons. This effect is called shielding effect.

The net nuclear charge experienced by valence electrons in the outermost shell is called the effective nuclear charge. It is approximated by the below mentioned equation.

$$Z_{eff} = Z - S$$



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